Accounting v2: measured check of the new ids and the versioned floor

aburan28/crypto PR #2465, branch `ecbench/fix-accounting`
Study `research/ecbench_accounting_v2_20261023`, run 2026-10-23
Binary built at head `7ed39a752` (sha256 `e265bbc0...54d50b1`)

What this is. An accounting and instrument correction check. It is not an algorithmic improvement and not a speed result: AGENTS.md's success criterion says relabelling and accounting changes are not improvements, and nothing here is reported as one. Counts are the result; wall time is descriptive only. Every figure is measured (recorded by `ecbench` in a session) or derived (the text says how). Preregistered in `PREREGISTRATION.md` before any data; amendments A1 to A4 are dated in the same file. Run report: `reports/run-20261023T0815Z/`.

1. Preregistered outcomes

id	outcome (preregistered)	result	verdict
O1	`rho.cubic_v2` at or above 1.0 of its floor on j = 0, interval excluding values below 1.0	S / floor = **1.419**, 95% interval **[1.132, 1.761]** on the registered j = 0 curve (n = 24; the `ecbench table` interval of S divided by the recorded `floor_s` 0.5117, derived). The first run on another j = 0 curve (archived, amendment A4): 1.514 [1.257, 1.800]	met on both curves
O2	strong v2 lowers mean S and set-up share against the frozen id at n = 13, 19, 37; ratio interval excluding 1	S ratio v2 / frozen 0.182 [0.171, 0.196], 0.223 [0.201, 0.248], 0.678 [0.499, 0.891]; set-up share 83.3% to 39.9%, 71.3% to 29.3%, 30.2% to 4.3%	met at all three sizes
O3 (a)-(c)	v2 minus frozen is exactly the verification phase; other phases equal; same recovered scalar	40 of 40 matched pairs for each generic method, 16 of 16 for `ic.large_prime`; A/A controls identical on every count	met
O3 (d)	verification phase `gae = 1.5 log2 r`	fails for every record: the phase is the exact double-and-add count of the planted scalar, a whole number of 19 to 29 GAE here (amendment A2)	**not met as written**; replacement (d') met 240 of 240, post hoc
O4	frozen ids' counts equal on the branch and on the main binary	504 of 504 matched records equal on every compared count field; on j = 0 only the floor figures differ, by design	met

Quoted figures against measured (all within the preregistered 15% band, so none is reported as differing). j = 0, the review's curve (first run, archived): 1.514 quoted, 1.514 measured; `rho.cubic` on the old floor 0.91 quoted, 0.911 measured. On the registered curve used for the primary run (a different curve, so the review's figures are not a prediction for it): 1.419 and 0.720. Strong reference frozen to v2, mean S and set-up share, quoted then measured: n = 13, 14.0 to 2.55 and 84% to 38% against 14.12 to 2.57 and 83% to 40%; n = 19, 2.95 to 0.68 and 77% to 31% against 3.16 to 0.705 and 71% to 29%; n = 37, 0.293 to 0.209 and 31% to 4.5% against 0.291 to 0.197 and 30% to 4.3%. The scratch runs used other seeds and targets; the largest differences are n = 19 frozen S (+7%), its share (-8%) and n = 37 v2 S (-6%).

2. Old id to new id

frozen id (counts unchanged)	v2 id	what changes in the count
`rho.cubic`, `rho.quartic`	`rho.cubic_v2`, `rho.quartic_v2`	the `[kappa]G` set-up ledger is inside `total_gae` (phase `automorphism_setup`); the walk of `rho.negation_v2`
`rho.negation`, `rho.signed_frobenius`	`rho.negation_v2`, `rho.signed_frobenius_v2`	Brent cycle detection beside the 16-point window; lost-walk steps and table queries and inserts counted
`rho.signed_frobenius_strong`	`rho.signed_frobenius_strong_v2`	lanes, distinguished-point bits and jump table scaled to r; start multiples charged; verification phase; spent work kept on exhaustion
`bsgs.textbook`, `bsgs.interleaved`, `bsgs.negation`	`bsgs.*_v2`	one charged final `[d]G = Q` (phase `verification`)
`kangaroo.vow`	`kangaroo.vow_v2`	the final check; restarts in their own phase `restarts`
`claw.pair_table`	`claw.pair_table_v2`	the final check; the `recover` phase inside the `max_adds` budget
`ic.large_prime`	`ic.large_prime_v2`	the `[r]Q` membership check and the final check counted in `verification`
floor of every record	`boundaries.floor_rule = "automorphisms/v2"`	A = 6 on j = 0 (p = 1 mod 3), 4 on j = 1728 (p = 1 mod 4); a record without the field is recomputed under the first rule (A = 2)

`rho.quartic_v2` and the j = 1728 floor were not run: the PR's specs contain no j = 1728 curve.

3. Q1: the j = 0 floor

Primary run, on the registered j = 0 curve `icv1-fp25-t9734-bcfe9f51` ($y^2 = x^3 + 1$ over $F_{23736841}$, group order 23 727 108 = 12 times 1 977 259, r = 1 977 259, `registered=true`, A = 6). Session `j0-v2` (the PR's spec `check-prime-j0-floor-v2.json`, which names this curve; 6 targets, 4 rounds, 24 runs per arm, all verified) and `j0-frozen-main` (same curve and seeds, frozen ids, recorded by the main binary). The first run used another j = 0 curve ($y^2 = x^3 + 2$ over $F_{4194397}$, r = 1 048 627) that is not in `docs/curves/registry.json`, which CI's curve-name check requires; the curve was switched (amendment A4) and the first run kept, unedited, in `archives/fp23-first-run/` as a replication (right panel of figure 1, table below).

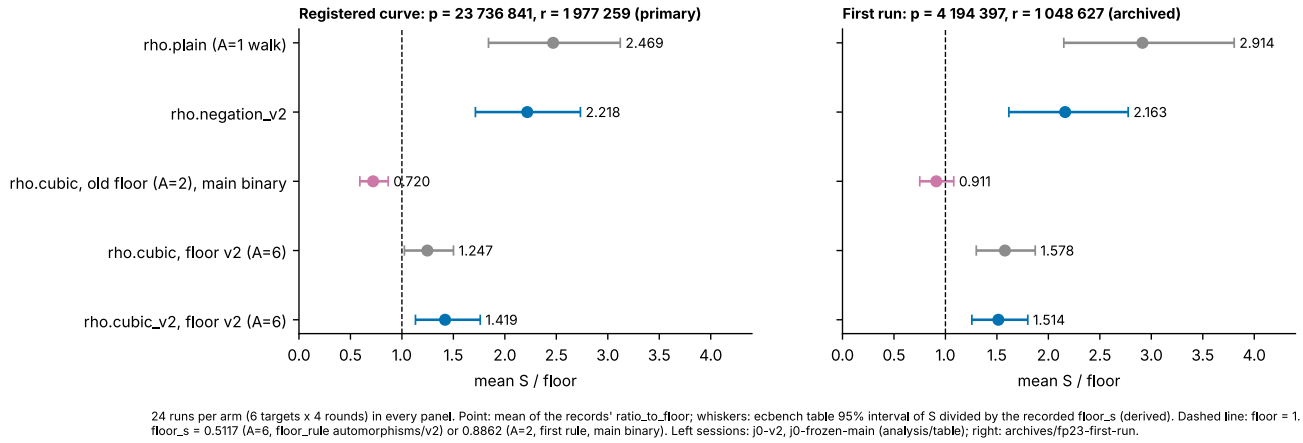


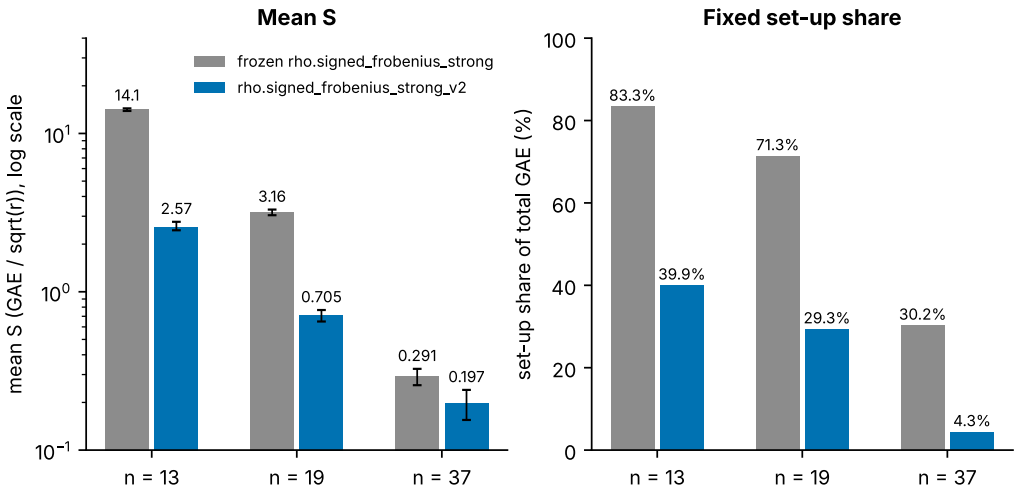
Figure 1: Mean S / floor on the j = 0 curves (dashed line: floor = 1). Left: sessions `j0-v2` ECBS1h0c64bbd17533 and `j0-frozen-main` ECBS1h7b9bf9250cf7. Right: the archived first run, ECBS1h70fa091addfe and ECBS1h1c109f746e68. 24 runs per arm.

arm (binary)	floor_s (A)	mean S	95% interval of S	S / floor	S / floor interval (derived)	first run (archived): mean S, S / floor	n
`rho.cubic_v2` (branch)	0.5117 (6)	0.7262	[0.579, 0.901]	**1.419**	[1.132, 1.761]	0.7746, 1.514 [1.257, 1.800]	24
`rho.cubic` (branch)	0.5117 (6)	0.6381	[0.524, 0.768]	1.247	[1.024, 1.501]	0.8074, 1.578	24
`rho.cubic` (main binary, first rule)	0.8862 (2)	0.6381	[0.524, 0.768]	**0.720**	[0.591, 0.867]	0.8074, 0.911 [0.750, 1.081]	24
`rho.negation_v2` (reference)	0.5117 (6)	1.1349	[0.877, 1.399]	2.218	[1.714, 2.734]	1.1067, 2.163	24
`rho.plain`	0.5117 (6)	1.2632	[0.942, 1.597]	2.469	[1.841, 3.121]	1.4909, 2.914	24

- **O1 is met on both curves:** `rho.cubic_v2` reads 1.419 of its floor with the interval above 1 (1.132 to 1.761), and 1.514 (1.257 to 1.800) on the first curve. The review's 1.514 was this first curve; on the registered curve the same method reads lower, which is a property of the curve and targets, not of the floor rule.
- The frozen `rho.cubic` counts are identical on the two binaries (72 of 72 records, both curves); only the floor differs. Under the first rule it reads 0.720 on the registered curve (interval [0.591, 0.867], excluding 1: a method that uses six automorphisms read below a floor computed with two) and 0.911 on the first curve (interval [0.750, 1.081], including 1); under `automorphisms/v2` it reads 1.247 and 1.578.
- Paired ratio `rho.cubic_v2` over `rho.cubic` of total S: **1.138 [0.899, 1.442]** (24 pairs) on the registered curve and 0.960 [0.826, 1.042] on the first. Both intervals include 1: this study cannot say whether charging the set-up ledger (about 26 GAE, 3% of S at these sizes, per the README) raises the total, since the v2 walk also differs (Brent cycle detection) and the two effects are not separated here.
- Paired `rho.cubic_v2` over `rho.negation_v2` is 0.640 [0.480, 0.853], so negation over cubic is 1.56 [1.17, 2.08] (inverted, derived) on the registered curve and 1.43 [0.96, 1.98] on the first; the prior $\sqrt{3} = 1.73$ is inside both intervals.

4. Q2: the strong reference, frozen and v2

Koblitz a = 0, public targets, same workloads and the same per-(workload, round) algorithm seed for both ids. Sessions `claim-n13` and `claim-n19` (committed claim specs, unchanged: 8 targets, 3 rounds, 24 runs per arm) and `strong-n37` (new spec, rho arms only: 8 targets, 4 rounds, 32 runs).



Koblitz a=0; 24 runs/arm (n=13, 19: 8 targets x 3 rounds) and 32 (n=37: 8 × 4). Bars: mean over runs, whiskers: ecbench table 95% interval.
Share = sum(setup + automorphism_setup phase GAE) / sum(total GAE) over the measured runs, from the records' phases (derived, same formula for both ids).

Figure 2: Strong reference, frozen against v2: mean S (log scale) and fixed set-up share. Sessions `claim-n13` ECBS1h0417f722e886, `claim-n19` ECBS1h7f68751e5578, `strong-n37` ECBS1h61cd8e1c1709.

n	log2 r	A	arm	mean S	95% interval	S / floor	set-up share (sum / sum)	(mean of runs)
13	10.97	26	frozen	14.118	[13.871, 14.405]	57.4	83.3%	83.4%
13	10.97	26	v2	2.571	[2.446, 2.767]	10.5	39.9%	40.2%
19	17.00	38	frozen	3.163	[3.045, 3.306]	15.6	71.3%	71.7%
19	17.00	38	v2	0.705	[0.649, 0.767]	3.47	29.3%	30.0%
37	27.78	74	frozen	0.291	[0.257, 0.326]	2.00	30.2%	32.0%
37	27.78	74	v2	0.197	[0.155, 0.240]	1.35	4.3%	5.2%

Paired ratio of total S, v2 over frozen (`ecbench compare`, cluster interval over workloads): n = 13 **0.182 [0.171, 0.196]**; n = 19 **0.223 [0.201, 0.248]**; n = 37 **0.678 [0.499, 0.891]**. The A/A control (v2 against its repeat) is 1.000 [1.000, 1.000] at all three sizes. Set-up share is computed from the records' `setup` plus `automorphism_setup` phases by one formula for both ids (derived); the frozen id records no `detail.setup_share`.

Reading it: the v2 reference is sized to r, so its fixed set-up is a smaller share and its mean S is lower at the same walk idea. That is the reference instrument being corrected, not a faster algorithm. At n = 13 and 19 the v2 reference is still 10.5 and 3.5 times its floor with 40% and 29% fixed set-up (above the 10% flag the README names), and every row is below the calibrated `log2 r` range for Koblitz curves (39). `ecbench table` on this head prints no flag column, although `methods::row_flags` exists (see limits).

5. Q3: one verification more

Same workload, round and algorithm seed for the frozen id and its v2 id. `pairs-prime`: 2^20-class prime curve `icv1-fp20-t727-cd198a38` (log2 r = 19.10); `pairs-koblitz`: Koblitz a = 1, n = 17 (log2 r = 16.00); `pairs-large-prime`: binary n = 11. Each pair has 40 matched records (8 targets, 4 rounds plus 1 warm-up); `ic.large_prime` has 16 (4 targets, 4 rounds).

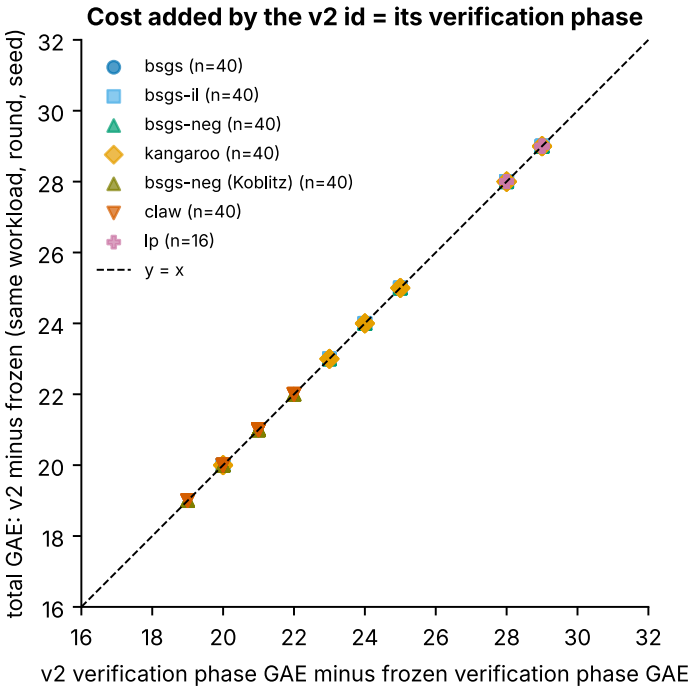


Figure 3: Added total GAE (v2 minus frozen, same workload, round and seed) against the v2 verification phase GAE, every matched record. Sessions `pairs-prime` ECBS1h5c19d0d7e485, `pairs-koblitz` ECBS1h001cebd5ff12, `pairs-large-prime` ECBS1h8ce582cbde15.

pair	matched	recovered, seed, status equal	other phases equal	delta total = delta verif.	one scalar mult	(d'): doubles, adds = bitlen - 1, popcount - 1	delta total (GAE)	S ratio v2 / frozen
`bsgs.textbook`	40	40	40	40	40	40	20-29	1.022 [1.021, 1.024]
`bsgs.interleaved`	40	40	40	40	40	40	20-29	1.028 [1.024, 1.036]
`bsgs.negation` (prime)	40	40	40	40	40	40	20-29	1.033 [1.030, 1.037]
`kangaroo.vow`	40	40	40 (setup = setup + restarts)	40	40	40	20-29	1.017 [1.014, 1.024]
`bsgs.negation` (Koblitz)	40	40	40	40	40	40	19-22	1.088 [1.073, 1.108]
`claw.pair_table`	40	40	40	40	40	40	19-22	1.069 [1.064, 1.075]
`ic.large_prime`	16	16	16	16	0 (two scalar mults)	n/a	28-29	1.0003 [1.0003, 1.0003]
A/A `negation_v2`	40	40	40	40 (delta 0)	n/a	n/a	0	1.000
A/A `frobenius_v2`	40	40	40	40 (delta 0)	n/a	n/a	0	1.000

- (a) to (c) hold for every matched pair. The positive control is the bsgs pairs themselves (the comparison sees a difference of 19 to 29 GAE); the negative control is the A/A rows (it sees none).
- (d) as preregistered fails for every record. The phase is `doubles = bitlength(d) - 1`, `adds = popcount(d) - 1` of the planted scalar d (40 of 40 for each of the six generic pairs), so its size follows d, not $1.5 \log_2 r$. Over the 8 targets its mean is 0.899 (prime) and 0.859 (Koblitz) of $1.5 \log_2 r$, a small sample of scalars. The README's "about 1.5 log2 r" is right in that sense; "exactly" was an over-reading (amendment A2).
- `ic.large_prime_v2`: the verification phase adds 28 to 29 GAE over the frozen one, two scalar multiplications (`[r]Q` and `[d]G`, about 14 GAE each at r = 991, derived from the 28 to 29 total) where the frozen verification phase held 90 adds of candidate checks and no scalar multiplication.
- The claw budget change (the `recover` phase inside `max_adds`) changed no run here: every matched claw pair differs only by the verification phase.
- The v2 ids cost 1.7% to 8.8% more S on these curves because a fixed 19 to 29 GAE sits on totals of 235 to 1500 GAE. This is a sizing statement for these curves, not for larger r (one scalar multiplication, O(log r), against O(sqrt r)).

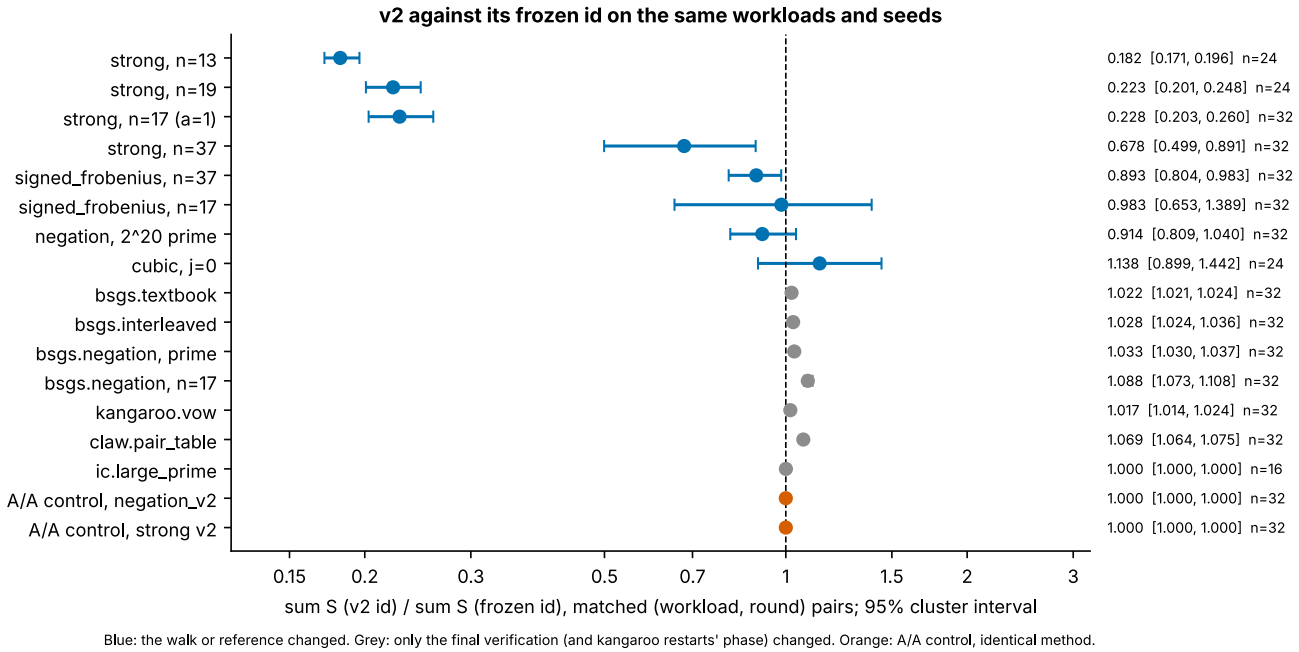


Figure 4: Total S of each v2 id over its frozen id on matched (workload, round) pairs, 95% cluster interval from `ecbench compare`, with the A/A controls. Sessions as in figures 2 and 3, `j0-v2` and `strong-n37`.

6. Q4: frozen ids against the main binary

The same frozen-only spec files were run once on the branch binary and once on the main binary (`ecbench-base`, sha256 `a59c4014...f81f`; the coordinator's build of main `3a1ea541d`, whose source tree and build log are not in the repository, and which does not embed its commit: this provenance is the coordinator's word). Compared per (arm, workload, round, warm-up): status, recovered scalar, algorithm seed, `total_gae`, unpriced counters, every phase's adds, doubles, scalar_mults, gae and native counts, all counters, and the run id.

spec (arms)	matched	all count fields equal	floor figures equal	`compare --b-dir` ratio
frozen-prime (`rho.negation`, `rho.plain`, `bsgs` x 3, `kangaroo.vow`)	240	**240**	240	1.000 [1.000, 1.000] each arm
frozen-koblitz (`rho.signed_frobenius`, strong, `claw.pair_table`, `bsgs.negation`)	160	**160**	160	1.000 each arm
frozen-large-prime (`rho.signed_frobenius`, `ic.large_prime`)	32	**32**	32	1.000 each arm
frozen-j0 (`rho.negation`, `rho.cubic`, `rho.plain`)	72	**72**	0 (A = 2 on main, A = 6 on branch, by design)	1.000 each arm

Total **504 of 504**. A second check replays committed sessions whose methods are frozen ids on the branch binary (`audits/committed/`): `ecbench_calibration_20261002` `followup-prime` (`kangaroo.vow`, `rho.negation`) and `followup-koblitz` (`rho.plain`, `rho.signed_frobenius`), 24 replays each; `ecbench_numa_vm_20261002/auto` (`bsgs.negation`, `kangaroo.vow`, `rho.negation`), all 48; `ghs_curve_comparison_20261009` `panel-01` and `non-koblitz-01`, 24 each; `ecbench_wide_gf2m_20261004/smoke-m83` (`claw.pair_table`, strong), all 4; `ecbench_bounds_challenges_20261006/koblitz-toy-reference-e1` (strong), 24; `ecbench_factor_bases_20261013/koblitz17`, all 20; `ecbench_n37_shared_rank_20261004/mac_arm64_l0_01`, all 15. Every replay reproduced; 0 problems.

Two committed sessions fail a replay on the branch binary and fail it identically on the main binary under `--replay-all` (96 and 16 problems, only their `ic.pipeline` root-finding and inherited-F4 arms): `ecbench_rootfind_engine_20261012` `claim-n13` and `smoke-koblitz`. This is a pre-existing condition, not caused by the PR (receipts `audits/committed/PREEXISTING_*`); CI's PR audit uses `--replay 0`, which passes.

7. Exploratory results and the pricing band

- **Brent cycle detection changes totals.** `rho.negation_v2` over `rho.negation`: 0.914 [0.809, 1.040] (2^20 prime, 32 pairs). `rho.signed_frobenius_v2` over `rho.signed_frobenius`: 0.983 [0.653, 1.389] (n = 17) and **0.893 [0.804, 0.983]** (n = 37). Runs with a recorded cycle longer than the window: 30 of 32 (`rho-neg-v2`, prime), 13 of 24 (`rho.cubic_v2`, registered j = 0 curve; 12 of 24 on the first curve), 10 of 32 and 5 of 32 (`signed_frobenius_v2`, n = 17 and 37). The README's "mean rose 14%" at n =

37 is the opposite direction to these paired totals (v2 lower); the README cites the review's runs (n = 80, with capped walks), and this study does not resolve the difference. Not a preregistered test.

- **Pricing band (AGENTS.md 5a)**, `analysis/band/`: every v2 arm emits its uncharged counters (the `uncounted` column is empty for every arm), so all arms are ranked across 0, 0.1 and 1 GAE per native operation. At n = 37 strong v2 is below the frozen strong across the band (S 0.197, 0.234, 0.560 against 0.291, 0.331, 0.691); strong v2 and `rho.signed_frobenius` are unresolved there.
- **Distance to the matched rho (success criterion).** The index-calculus arms of the committed claim specs are unchanged by this PR; the reference they are read against is what changed (`analysis/gap_to_rho.tsv`, band weights 0, 0.1, 1). The best IC arm across the band, `ic-rootfind-m2`, is 10.6 to 11.0 times strong v2 at n = 13 (2.0 to 2.3 times the frozen strong) and 34 to 49 times at n = 19 (10 to 11 times the frozen strong). The corrected references make the measured gap larger, not smaller. These are distances remaining, not a gate.

8. Deviations, limits, not done

- Amendments (dated in `PREREGISTRATION.md`): A1 added the frozen `ic.large_prime` cross-binary spec; A2 records that outcome (d) was too strict and adds the post hoc (d'); A3 records that the main-binary j = 0 session fails the branch audit by design; A4 records the switch of the j = 0 curve to a registered one, with the first run kept as a replication.
- `j0-frozen-main` (registered curve) is a verbatim archive (`archive/j0-frozen-main.tar.gz`, sha256 beside it; the first curve's is `archives/fp23-first-run/j0-frozen-main.tar.gz`), not a session directory: the branch binary's audit rejects it ("used 6 automorphisms, the curve has 2", 25 problems in both runs), which is the gap the floor rule closes; the main binary accepts it. Committed as a session it would fail CI's re-audit of every `ecbench.session/v1`.
- Sessions record the worktree's `git_commit` at run time (`c5ca4380e`, `2833cfc7a` and, for the registered-curve j = 0 sessions, `3d310548d`, above the binary's source head `7ed39a752`; the commits between add only this study). The binary sha256 is the same in every branch session.
- Isolation: L1 or better only, not the L2 the specs request on most records (a shared 4-vCPU virtual machine with other agents building). Counts are unaffected; no wall-time statement is made, and median solves (0.1 to 1.5 ms) are below the harness's 1 ms admission minimum.
- `ecbench table` prints no flag column on this head (`methods::row_flags` and `setup_share` are referenced only by unit tests); the README of this PR now says the column is a follow-up (the share is in `detail.setup_share`). The flags above are computed by hand from the same thresholds.
- Not run: `rho.quartic_v2` and the j = 1728 floor; sizes between or beyond n = 13, 19, 37 for the strong reference (no scaling claim); a replay of the branch's sessions on the main binary.
- Curve names: the first j = 0 curve is not registered, so its sessions, tables and the PR's original spec text live only under `archives/` (exempt from `scripts/check_curve_names.py`); a run of that script in this sparse checkout (files absent from it read as empty) reports 0 problems, both plain and with `--diff origin/main`.
- Sessions are 16 directories of about 16 MB (the per-record isolation capsule dominates; the whole study directory is 20 MB), more than the few MB asked for; git compresses `records.jsonl` roughly tenfold.
- No `benchmarks/` entry: nothing here measures how fast an algorithm runs or reports a speedup, and no validation-gate row is created. No leaderboard change: no row of `docs/ic/leaderboard.json` is added or moved (accounting changes are not improvements). Canonical graphs checked and left unchanged: `docs/ic-leaderboard.html`, `docs/ic/LEADERBOARD.md`, `docs/ic/leaderboard.json`, `docs/index-calculus-scoreboard.html`, `docs/bounds/*`; they cite frozen ids, whose counts Q4 shows unchanged. A follow-up may add v2-reference columns.

9. Files and reproduction

`PREREGISTRATION.md`, `SPEC_SHA256.txt`, `specs/`; `sessions/` (never edited), `archive/`, `archives/fp23-first-run/` (the first j = 0 run: sessions, archive, specs, audits, analysis), `audits/` (replay-all receipts beside the sessions, never inside), `logs/`, `host.txt`; `analysis/` (commands `run.sh`, `run_a1.sh`, `audit.sh`, `replay_committed.sh`; jq and shell scripts that produce every number; `compare/`, `table/`, `band/`, `summary/`, `pair_check.json`, `cross_binary.json`, `gap_to_rho.tsv`, `inventory.tsv`; `plot.py`, which plots recorded values only); `figures/` (SVG, 150 dpi PNG and `*.data.json` with the numbers drawn); `REPORT.typ`, `REPORT.pdf`; `reports/run-20261023T0815Z/`.